

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		rate = $k[\text{CH}_3][\text{S}_2\text{O}_3^{2-}]$ award (1) for rate equation containing $k[\text{CH}_3\text{I}]$ award (1) for second order rate equation		1	1	2		
		(ii)		change E_a to 81100 J mol^{-1} (1) rearrange equation to $T = -\frac{Ea}{\left(R \ln \frac{k}{A}\right)}$ (1) $T = 346 \text{ K}$ (1)	1		1 1	 3	 3	
	(b)	(i)		quenching is the sudden stopping / significant slowing of a chemical reaction in a sample (1) ensure sample composition does not change between taking sample and analysis OR during analysis (1)	1 1			2		2
		(ii)		starch	1			1		1
		(iii)		canary/bright yellow	1			1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	I	catalyst in the same physical state as reaction	1			1		
			II	award (1) each for any two of following - transition metals have empty orbitals so can form bonds to reactant molecules / form complex - transition metals have variable oxidation states (so they can oxidise/reduce the reactants) - products are released and the metal returns to the original oxidation state	2			2		
		(ii)		$[H^+]^2 = K_a \times [\text{acid}]$ $[H^+] = 1.97 \times 10^{-3} \text{ mol dm}^{-3}$ $\text{pH} = 2.7$		3		3	3	
				Question 10 total	8	4	3	15	6	4